COMP662 — Generative Artificial Intelligence

Assignment 1: Synthetic Dataset Generation

Semester 2, 2026 | Weight: 20% of final grade | Individual Assessment

Item	Detail
Due Date	5:00 p.m, Sunday, 23 August 2026
Submission Platform	Akoraka | Learn
Assessment Type	Individual
GenAI Use	Permitted (see Section: GenAI Policy)

Late Submission Policy

Time Late	Penalty
Less than 6 hours late	5% late penalty
Between 6 and 24 hours late (up to 1 day)	12.5% late penalty
Between 24 and 48 hours late (1-2 days)	25% late penalty
Over 48 hours late (2 days or longer)	100% late penalty

Penalties are applied to the mark achieved, not the maximum available. Extensions more than 3 days require formal aegrotat application with supporting documentation before the due date.

Academic Integrity

This is an individual assessment. Each student must produce their own analysis and code. While general concepts may be discussed verbally with peers, all submitted work (including code, visualisations, and written interpretations) must be entirely your own. Submissions containing duplicated or substantially identical content will result in zero marks for all parties involved.

You must not share your code, notebook, or any written material with other students. Allowing another student to copy your work is also a breach of [academic integrity].

GenAI Policy

Generative AI (GenAI) tools are permitted in this course. You may use them for brainstorming, code debugging, writing assistance, and conceptual clarification. However, you remain fully responsible for the correctness and originality of all submitted work.

You must include a GenAI Acknowledgement file stating which tools you used and for what purpose. If no GenAI tools were used, state this explicitly. Failure to declare GenAI use where it has been used may be treated as a breach of academic integrity.

Overview

You will be provided with a real-world bean dataset containing dimension-based and shape-based morphological variables that describe the form, shape, and structure of individual beans, together with bean variety class labels. You will first establish a baseline machine learning classifier using the original dataset, then train a generative model to synthesise additional data, and critically evaluate the quality, diversity, and usefulness of the generated samples using both statistical measures and classification performance.

The goal of this assignment is to investigate whether synthetic data generated by a generative model can effectively augment the bean dataset and improve bean classification performance. Rather than pursuing the highest possible classification accuracy, the emphasis is on selecting and justifying an appropriate generative approach, critically evaluating the realism and usefulness of the generated data using multiple evaluation metrics, and communicating your methodology and findings in a clear, evidence-based manner.

Learning Objectives

Upon completing this assignment, you should be able to:

- Prepare a real-world tabular dataset using appropriate preprocessing techniques and justify the choices made.
- Design, implement, and train an appropriate **generativ**e model to synthesise realistic tabular data, justifying your modelling choices.
- Evaluate the fidelity, diversity, and utility of synthetic data using appropriate statistical measures and classification performance.
- Develop and compare baseline and synthetic-data-augmented **classification** models, justifying the design of your classification models.
- Analyse and communicate your findings clearly in the notebook.

Dataset

You are provided with one real-world dataset file for this assignment:

- **Filename:** train.csv

This dataset contains dimension-based and shape-based morphological measurements derived from bean grains collected in Turkey. The data are based on high-resolution camera images of grains of registered bean varieties. Each observation corresponds to a single bean grain. The task is to predict the bean variety based on these measurements. Class is the target variable and takes integer values from 0 to 4. These values should be treated as five distinct bean categories. The goal of the classification task is to predict exactly one of five dry bean varieties for each grain. The dataset has been modified from the source to better suit your assessment. In the original dataset, bean categories include Seker, Barbunya, Bombay, Cali, Dermosan, Horoz, and Sira.

Property	Value
Number of observations	12,111
Number of input features	10
Target variable	Bean Category
Number of classes	5 (integer values 0 to 4)

Feature Descriptions:

Feature	Description	Type
Area (A)	The area of a bean and the number of pixels within its boundaries.	Int
Perimeter (P)	Bean circumference is defined as the length of its border.	Continuous
Major axis length (L)	The distance between the ends of the longest line that can be drawn from a bean.	Continuous
Minor axis length (l)	The longest line that can be drawn from the bean while standing perpendicular to the main axis.	Continuous
Eccentricity (Ec)	Eccentricity of the ellipse having the same moments as the region.	Continuous
Convex area (C)	Number of pixels in the smallest convex polygon that can contain the area of a bean.	Int
Extent (Ex)	The ratio of the pixels in the bounding box to the bean area.	Continuous
Solidity (S)	Also known as convexity. The ratio of the pixels in the convex shell to those found in beans.	Continuous
Roundness (R)	Calculated with the following formula: (4piA)/(P^2).	Continuous
Compactness (CO)	Measures the roundness of an object: Ed/L.	Continuous
Class	Bean Category.	Int

Note: The class distribution may not be balanced. You should examine the distribution of Class and consider its implications for preprocessing, metric selection, classifier training, and generative modelling. The dimension-based features indicate absolute physical size and boundary lengths of each bean grain. The shape-based features describe geometric properties, curvature, and proportions independent of scale, characterizing the overall morphological structure of the grain.

Instructions for loading the dataset:

- Download train.csv from Akoraka | Learn and place it in your working directory.
- Load the dataset programmatically using Python.

Tasks

This assignment consists of six tasks. All your written analysis should be reported in the Python notebook along with your code.

Task 1: Dataset Understanding and Concise Exploratory Data Analysis (10 marks)

Begin by exploring the dataset to understand its structure and identify key patterns. You are not required to analyse every feature in exhaustive detail. Instead, focus on providing a clear, concise overview and highlighting the most relevant findings. You should give your answer in the Python notebook.

You should:

- Report the class distribution of Class using both counts and percentages.
- Select 3-5 representative or potentially important features for deeper analysis with visualisation.
- Provide one correlation analysis or correlation heatmap for continuous features.
- Summarise how the Exploratory Data Analysis (EDA) findings may affect preprocessing, model choice, metric choice, or cross-validation design.

Task 2: Baseline Classifier Construction (10 marks)

Based on your EDA, perform any necessary data preprocessing and build a baseline machine learning model. Your choice of evaluation strategy is critical and must be justified with reference to the imbalanced nature of the dataset.

You should:

- Perform and justify necessary data preprocessing steps (e.g., feature scaling, encoding).
- Select a suitable classification model and briefly justify your choice.
- Select a primary evaluation metric, a supporting evaluation metric, and an appropriate cross-validation strategy. Justify all of them.
- Train and evaluate your baseline classifier using the validation strategy you selected. You should also create your own held-out test dataset. Ensure that all preprocessing is performed within your training data only to avoid data leakage.

Clearly state what you chose to do, what you chose not to do, and why. Unjustified data preprocessing steps will not receive full marks. Use cross-validation on the training set for model development and hyperparameter selection (if applicable), and reserve the held-out test set for the final evaluation only.

Task 3: Generative Model Development and Justification (20 marks)

Develop a Generative AI model capable of modelling tabular data. This task focuses on your understanding of generative modelling.

You should:

- State the generative model you have chosen and justify your choice.
- Describe the architecture of the chosen generative model, including its main components and the role of each component in learning and generating synthetic tabular data.
- Explain how the model represents and generates tabular data, including how class labels (or other conditioning information) are incorporated where appropriate.
- Describe any data preprocessing or transformations required for the generative model.
- Discuss any assumptions, limitations, or potential risks associated with your chosen model.

Task 4: Data Synthesis and Quality Assurance (25 marks)

Train the generative model on your training dataset and synthesise a new batch of artificial data. You must demonstrate that the generated synthetic data is statistically realistic and consistent with the characteristics of the original dataset.

Evaluation Metrics:

You should:

- Describe the training process of the generative model.
- Explain and justify your data generation strategy, including the number of synthetic samples generated and whether the synthetic data is intended to augment specific classes or the whole training data.
- Generate a synthetic dataset using your trained generative model.
- Compare the real and synthetic datasets using appropriate visualisations and statistical analyses (e.g., feature distributions, correlation matrices, class distributions, or summary statistics).
- Critically evaluate the quality of the generated data, discussing both strengths and limitations, supported by appropriate metrics and evidence.

Task 5: Classification with Synthetic Data and Performance Analysis (25 marks)

Use the synthetic data generated in Task 4 to augment the original training dataset and investigate whether synthetic data improves classification performance.

You should:

- Utilise the synthetic data to develop an augmented classifier by retraining the same classification model using the **same preprocessing pipeline** and **hyperparameters** as the baseline model developed in Task 2. No additional tuning should be performed in your augmented classifier.
- Evaluate both the baseline and augmented classifiers. Compare the primary and supporting evaluation metrics for both models. The held-out test set should be the same.
- Provide a class-level performance analysis (e.g., confusion matrix, per-class precision/recall/F1) to examine the effect on minority and majority classes.
- Critically discuss whether the synthetic data improved, degraded, or had little effect on classification performance, and explain possible reasons for the observed outcomes.

Task 6: Hidden Test (10 marks)

Submit the final classification model that you believe provides the best generalised classification performance. You may submit either your baseline or augmented classification model. You need to give instructions on how to run your submitted classification model on a new CSV file and generate bean category predictions. Models will be evaluated on a hidden test set. Marks will be awarded based on the model's ranking among all submissions as follows:

- Top 10%: 10 marks
- Top 20%: 9 marks
- Top 30%: 8 marks
- Top 40%: 7 marks
- Top 50%: 6 marks
- Remaining submissions: 5 marks, provided the model achieves a minimum performance threshold determined by the examiner.

Models that do not meet the minimum performance threshold will receive fewer than 5 marks, with marks awarded at the examiner's discretion based on performance. If your submitted model cannot be executed, you will receive minimum marks from this task.

Submission Requirements

Create a GitHub repository to store all your submission documents. Submit a text file named GitHub_URL.txt containing only the repository URL. Name your repository StudentID-COMP662A1, replacing StudentID with your student ID (e.g., 12345678-COMP662A1). The GitHub repository should contain the following documents:

File	Description
StudentID_Assignment1.ipynb	Jupyter Notebook (or Python scripts) containing all training and testing code, analysis, and visualisations.
Saved final classification model	Save your final classification model in a format of your choice. It must be loadable and executable in Python.
preprocessing.py (optional)	This submission is optional. You only need to submit it if your model depends on your customised preprocessing steps. This program will be used to preprocess the unseen test data before running the submitted classification model.
run.txt	A brief description of how to run your submitted model. Also describe how to use your preprocessing.py file if you submit it. Your instructions should explain how to use the model to evaluate a new CSV file similar to your training dataset and generate bean category predictions. You should list installed packages and their versions.
GenAI_Acknowledgement.txt	A short statement of which GenAI tools were used and for what purpose, or a statement that no GenAI tools were used.

Ensure that the notebook and model use relative file paths and can be run in a clean Python environment. Remove unnecessary debugging output, but retain the outputs required to assess your results. You **must not** push additional commits or modify the repository after the assignment deadline. Otherwise, a late penalty will apply.

Marking Criteria

The following table outlines the marks allocated to each task and provides general descriptors for three performance bands.

Task	Marks	High (80-100%)	Adequate (50-79%)	Low (0-49%)
Task 1: Dataset Understanding & Concise EDA	10	Clear dataset and feature analysis; appropriate concise visualisations and correlation analysis; meaningful interpretation linking EDA findings to downstream modelling, metric, and Cross-Validation (CV) choices.	Basic dataset and feature analysis with visualisations present; some correlation analysis; limited interpretation connecting findings to downstream modelling decisions.	Incomplete or incorrect dataset and feature analysis; visualisations missing or unclear; little to no interpretation or correlation analysis.
Task 2: Baseline Classifier Construction	10	All decisions (preprocessing, model choice, metrics, CV) thoroughly justified, specifically addressing the imbalanced nature of the data; clear reasoning for what was and was not done.	Appropriate model and baseline executed; some preprocessing performed with partial justification; basic understanding of metric/CV choices for imbalanced data demonstrated.	Minimal or unjustified preprocessing; inappropriate classification algorithm or metric choices; CV missing or incorrectly implemented.
Task 3: Generative Model Development and Justification	20	Generative architecture clearly explained and suitable for tabular data; thorough justification provided; deep, critical discussion of assumptions, limitations, and risks; Detailed description of any preprocessing steps.	Generative model stated with basic architectural explanation; partial justification provided; some discussion of limitations/risks; partial description of preprocessing steps.	Single generative model chosen with little/no justification; lacks architectural understanding; missing discussion on limitations; preprocessing steps are used but not described.

Task 4: Data Synthesis and Quality Assurance	25	Clear training process for the generative model; clear, well-justified generation strategy; highly effective visual and statistical quality checks (e.g., feature distributions, correlation matrices); comprehensive evaluation of synthetic data quality.	Some description of the training process for the generative model; synthetic data generated using a partially justified generation strategy; adequate visual/statistical quality checks provided, but lacks depth; basic evaluation of data quality.	Training process for the generative model unclear or missing; Generation strategy unclear or missing; synthetic data quality not evaluated or checks are entirely inappropriate; generated data is of poor quality.
Task 5: Classification with Synthetic Data and Performance Analysis	25	Baseline algorithm and hyperparameters correctly reused; Comprehensive comparison of primary/secondary metrics; clear class-level breakdown; deeply insightful and critical analysis of exactly why the synthetic data impacted the results the way it did.	Minor discrepancies in hyperparameter consistency or baseline reproduction; Basic comparison of metrics between baseline and augmented models; some class-level observation; basic discussion of the synthetic data's impact but lacks deep critical analysis.	Entirely different model or parameters used without reason; Superficial or missing comparison; no class-level breakdown provided; fails to discuss or analyse why the outcome occurred.
Task 6: Hidden Test	10	Top 30% performance on the hidden test set.	Top 50% performance on the hidden test set.	Performance not in the top 50%; Fail to pass the minimum performance threshold; Model is not loadable or executable in Python.
Total	100			

Additional Guidance

- The focus is on understanding and justification, not on achieving the highest possible accuracy. A well-justified simple model with thorough analysis will score higher than a poorly justified complex model.
- If you tune hyperparameters, clearly separate the data or folds used for tuning from those used for reporting performance. Nested cross-validation is optional.